Problems

1. The hemoglobin B gene (*Hb*) has a common allele (*A*) of a SNP (*rs334*) that encodes the *Hb*A form of (adult) hemoglobinand a rare allele (*T*) that encodes the sickling form of hemoglobin, *Hb*S. Among 571 residents of a village in Nigeria, 440 were *A/A* and 129 were *A/T,* and 2 were *T/T* individuals were observed. Use the $\chi 2$ test to determine whether these observed genotypic frequencies fit Hardy–Weinberg expectations.

对所测得的数据，根据哈代温伯格定律：

A 基因频率：p = (440 + 129×1/2) / (440+129+2) = 0.884

T 基因频率：q = (129×1/2 + 2) / (440+129+2) = 0.116

3、平衡群体基因型频率的计算：

按哈代温伯格定律公式：

$p^2 + 2pq + q^2 = 1$,得：

AA: p^2=0.884*0.884=0.781

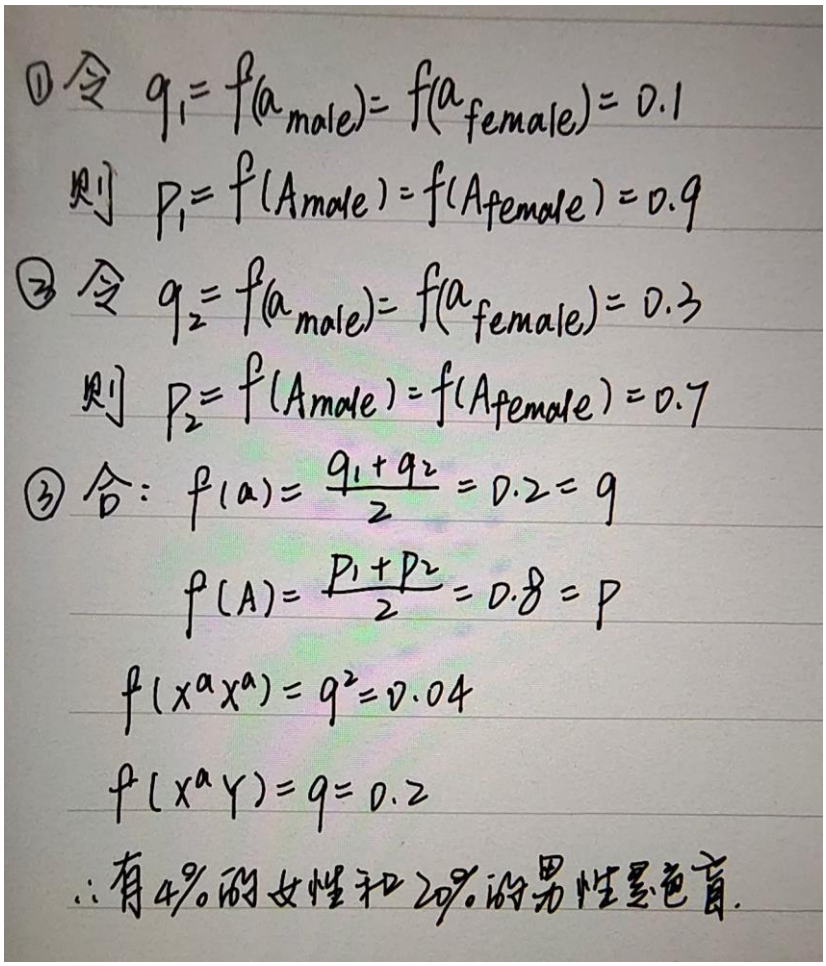
AT: $2pq$=2*0.884*0.116=0.206

TT: q^2=0.116*0.116=0.014

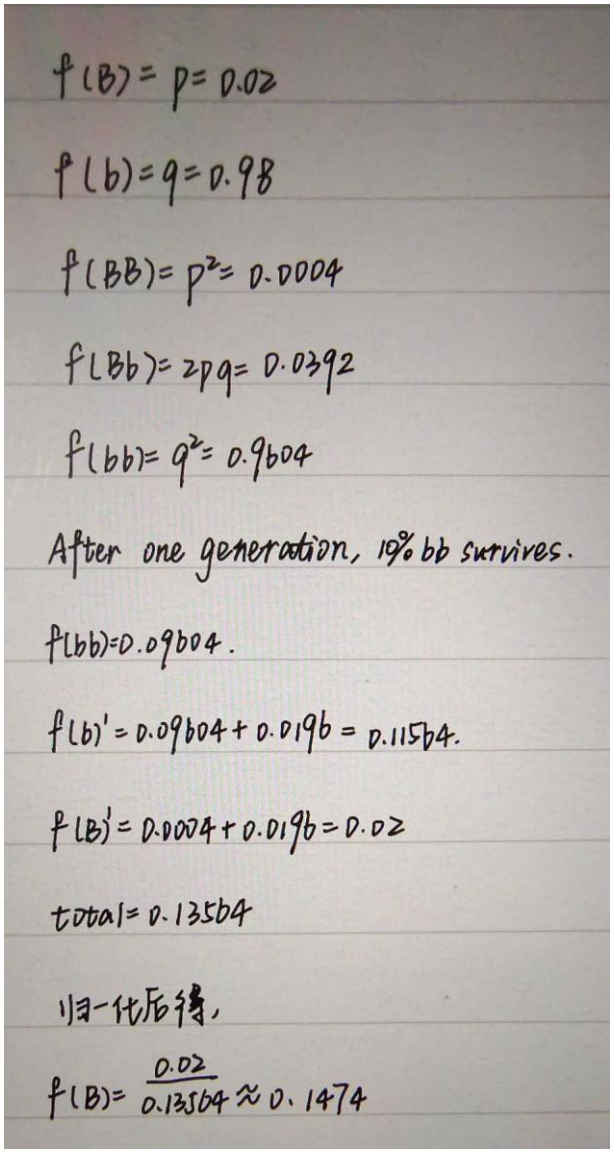
	AA	AT	TT	Total
实际频(O)	440	129	2	571
理论频(C)	(Np^2) 445.74	（N2pq） 117.51	（Nq^2） 7.74	N 571
$(O-C)^2/C$	0.074	1.123	4.261	5.459

通过查表，比较，发现当 df=1，χ^2 =5.459 时，p value 在 0.01 到 0.05 之间，差异不显著，不符合预期。

2. Color blindness in humans is caused by an X-linked recessive allele. Ten percent of the males of a large and randomly mating population are color-blind. A representative group of 1000 people from this population migrates to a South Pacific island, where there are already 1000 inhabitants and where 30 percent of the males are color-blind. Assuming that Hardy–Weinberg equilibrium applies throughout (in the two original populations before the migration and in the mixed population immediately after the migration), what fraction of males and females can be expected to be color-blind in the generation immediately after the arrival of the migrants?



3. A farmer plants transgenic Bt corn that is genetically modified to produce its own insecticide. Of the corn borer larvae feeding on these Bt crop plants, only 10 percent survive unless they have at least one copy of the dominant resistance allele *B* that confers resistance to the Bt insecticide. When the farmer first plants Bt corn, the frequency of the *B* resistance allele in the corn borer population is 0.02. What will be the frequency of the resistance allele after one generation of corn borers fed on Bt corn?



4. Consider a population in which the frequency of allele *A* is $p = 0.7$ and the frequency of allele *a* is $q = 0.3$, and where the alleles are codominant. What will be the allele frequencies after one generation if the following occurs?

(a) $w_{AA} = 1$, $w_{Aa} = 0.9$, $w_{aa} = 0.8$

(b) $w_{AA} = 0.8$, $w_{Aa} = 1$, $w_{aa} = 0.8$

